R2 — ML Research Track

Member: ______

Your role

You own the tracking sheet for the whole project. You run the ple arm (seed 0) and write the verification report.

Weekly tasks

Week 1 — Tracking sheet

- Task: Read README.md, RESULTS.md, and research/tinystories/README.md. Build the tracking sheet (Google Sheets). Columns: claim, published value, our value, status, owner.
- Deliverable: The tracking sheet link in the club Drive.
- Verify: Every published number in RESULTS.md has a row in the sheet.
- Learn: Neural network basics. Watch the 3Blue1Brown series on neural networks.

Week 2 — Launch ple seed 0

- Task: Launch the run: `uv run python -m research.tinystories.train --arm ple --vocab 32768 --steps 3000 --target-core 559000 --seed 0` on your Colab account.
- Deliverable: The run reaches 3000 steps. The checkpoint and the run JSON record land in `runs/`.
- Verify: Checkpoints save to Google Drive every 1000 steps.
- Learn: Embeddings and tokenization. Write a numpy lookup that maps token ids to embedding rows.

Week 3 — Collect ple results

- Task: Collect your two ple checkpoints. Record the val ppl for both seeds. Update the tracking sheet.
- Deliverable: The ple ppl numbers for both seeds.
- Verify: Your numbers sit within +/-0.05 of 11.41.
- Learn: The transformer block. Draw one block and label every tensor shape.

Week 4 — Verification report

- Task: Write the verification report. What matched? What did not? Why? Consider Colab precision, dataset slice drift, and seed differences.
- Deliverable: VERIFICATION_REPORT.md in the club repo.
- Verify: R1 reads it and agrees with every number.
- Learn: Loss and perplexity. Explain why lower ppl means a better model.

Week 5 — Improvise decision

- Task: Run the improvise decision session with the team. Capture the chosen direction and the domain for fine-tuning.
- Deliverable: The decision notes with the chosen direction.
- Verify: Every member can name the chosen direction.
- Learn: Attention math. Derive the softmax(QK^T)V formula by hand on one example.

Week 6 — Eval protocol

- Task: Run the eval protocol on the fine-tuned model. Collect samples and score coherence.
- Deliverable: The eval results table for the domain model.
- Verify: The protocol covers ppl and human coherence. Two members score the same text similarly.

- Learn: Fine-tuning. Read how a pretrained model adapts to a new domain.

Week 7 — Teach QKV math

- Task: Present the QKV attention math to the team at the meetup.
- Deliverable: Your teach-back slides.
- Verify: R1 can explain the shapes of Q, K, and V.
- Learn: Prepare your presentation deeply.

Week 8 — Track finalize

- Task: Close the tracking sheet. Mark every claim verified or failed with the evidence link.
- Deliverable: The closed tracking sheet.
- Verify: No row has an empty status.
- Learn: Write the "what I learned" page.

Learning path

1. Neural network basics: perceptron, MLP, gradient descent.
2. Embeddings, tokenization, next-token prediction.
3. Transformer block: attention, LayerNorm, residuals, MLP.
4. Loss, perplexity, sampling, generation.
5. Attention math, RoPE, KV cache. PLE paper.
6. Fine-tuning, transfer learning, PTQ.
7. Teach one topic to the team.
8. Write the "what I learned" page.

Resources

- Repo: https://github.com/slvDev/esp32-ai
- RESULTS.md: https://github.com/slvDev/esp32-ai/blob/main/RESULTS.md
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi
- TinyStories paper: https://arxiv.org/abs/2305.07759